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Cognition and Renewal: Comparing CEO and Organizational Effects on Incumbent Adaptation to Technical Change

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We investigate the conditions under which managerial cognition affects the timing of incumbent entry into a radical new technological market. We address this question using a longitudinal study of communications technology firms entering the fiber-optics product market. Using a hazard rate model, we investigate the relevance of cognition based on the direction of CEO attention. We find that attention toward the emerging technology and the affected industry is associated with faster entry, and attention to existing technologies is associated with slower progress. Second, we assess the extent to which the effect of cognition is dependent upon the levels of relevant organizational factors and find that CEO attention to the emerging technology may amplify the effects of industry orientation. Managerial cognition is important in understanding organizational outcomes, and considering both the direction of cognition and its interaction with organizational factors provides a more nuanced view of entry behavior. These results contribute to the literatures on incumbent response to technical change and new product development by suggesting that context-specific managerial cognition has a separate and important influence on the degree and direction of strategic renewal. We argue that managerial cognition is therefore a dynamic managerial capability that can shape adaptation by established firms.

Key words: innovation; market entry; cognition; attention; dynamic managerial capabilities; strategic renewal

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1. Introduction

Incumbent firm strategic reorientations are often triggered by a technical change that requires managers to choose whether or not to enter a new market, a choice that is often uncertain and risky. There have long been two distinct views of the role of top managers in such reorientations. One view suggests that managers matter only as seamless conduits of plans based on matches between organizational capabilities and environmental conditions. From this standpoint, new market entry choices are viewed as rational decisions based on the desire to avoid self-cannibalization (Henderson 1993), limit sunk costs (Williamson 1985), or impede rivals (Caves 1984). The presumption is that managers choose to pursue new opportunities when their firms have the proper capabilities to succeed in the new market (King and Tucci 2002, Klepper and Simons 2000, Lambkin 1988), and that their decision-making process is a relatively straightforward economic calculation. An alternative view suggests that managers could matter separately from the effects of the organization and the environment. Scholars using variance decomposition approaches (Adner and Helfat 2003, Thomas 1988, Wasserman et al. 2001) have long argued that this is the case in general. Tushman and colleagues (Tushman and Rosenkopf 1996, Virany et al. 1992) have

delved into these effects in the specific context of market turbulence to show that top management changes can enable effective organizational reorientations. Such findings imply that the connection between the external environment and the internal organization is problematic and therefore can be shaped by senior executives who define and legitimize particular courses of action.

Recently, scholars have begun to put forward resolutions to this conflict between the view of managers as essentially “rubber stamp” decision makers guided by organizational constraints and that of managers as independent agents with a separate effect on strategy in turbulent contexts. Adner and Helfat (2003) theorize that both organizational and managerial effects are crucial in understanding firm action in the face of environmental change, and more importantly, that it is the interaction between these two effects that makes the difference. They describe this interaction as a dynamic managerial capability to “build, integrate, and reconfigure organizational resources and competences” (Adner and Helfat 2003, p. 1012). One such dynamic managerial capability that they identify is managerial cognition. This idea is fully consistent with research in the managerial cognition tradition, which has shown that managerial attention is especially crucial for shaping strategic choice

and action when responses to an uncertain environment require costly, risky investments (Barr 1998, Cho and Hambrick 2006, Kaplan et al. 2003, Ocasio 1997). Less has been done empirically, however, to examine the relationship between these cognitive and organizational dynamics. Tripsas and Gavetti's (2000) case study of Polaroid's failure to respond to the digital revolution in photography demonstrated that both managerial and organizational factors were important in affecting the degree of firm reorientation. Polaroid's actions were constrained, not by a lack of organizational capabilities in digital technologies, but by managerial beliefs that were not aligned to the new market opportunity. These findings whet the appetite for more systematic studies of such effects across a greater number of firms. Larger-scale empirical work would provide the opportunity to understand better the conditions under which managerial cognition might make a difference in incumbent strategic renewal, and to examine the degree to which such cognition operates as a dynamic managerial capability in interacting with organizational factors.

With this goal in mind, this paper explores two conditions under which managerial cognition may have an impact on a specific type of strategic reorientation, that of entry into a new product market. Agarwal and Helfat (2009, p. 282) define strategic renewal as the "replacement of attributes of an organization that have the potential to substantially affect its long-term prospects," and the introduction of a new product that thrusts that firm into a completely new market would clearly serve as a vehicle for strategic renewal. We measure cognition in terms of the attention a CEO devotes to particular issues (Ocasio 1997). First, this paper evaluates the degree to which the direction of CEO attention affects organizational outcomes. We focus on the CEO's attention to three domains that have been shown to be relevant at the *organizational* level when new technologies emerge: the emerging technology, the existing technology, and the affected industry. Research has shown that firms that are more oriented toward technologies associated with the emerging regime may be better positioned to take advantage of it (Cattani 2005). However, orientation toward existing technologies might lead to competency traps that would disable effective firm response (Leonard-Barton 1992). Finally, industry orientation can have a beneficial effect if it creates relevant complementary assets that enable a firm to capitalize on new technologies (Klepper and Simons 2000, Tripsas 1997). We assess these three domains at the level of the CEO as well as of the organization so that we can understand the impact of the direction cognition even in the presence of organization-level effects. Second, we examine interactions between CEO attention and comparable organizational factors to gauge the degree to which the impact of cognition depends directly on the level of capabilities possessed by the organization.

To study these effects, we examine incumbent communications technology firms' response to the emergence of fiber optics over the past three decades. This context is an appealing setting to examine how CEO attention and organizational orientations could explain heterogeneity in outcomes. Entry into the fiber-optics market was costly and risky, and therefore deserving of CEO-level attention. Some incumbent firms entered quickly, some more slowly, and some not at all. To address the effect of CEO attention and organizational orientation, we focus on the classic question from the product introduction literature: explaining heterogeneity in the timing of entry into a new product market. Using a hazard rate model, we show that the effect of CEO attention depends on its direction: attention to the emerging technology and the affected industry is associated with accelerated entry, whereas attention to existing technologies is associated with slower progress. We also demonstrate that the impact of CEO attention depends on the organizational orientation of the firm, as the effect of attention to the emerging technology is amplified by increases in an organization's experience in the industry. These results contribute to the literature on managerial cognition by outlining circumstances under which managers have more or less effect on firm strategic reorientations, and to the literature on incumbent response to technical change by suggesting that CEO attention has an important influence on the behavior of organizations as they seek to renew themselves.

2. Theory and Hypotheses

Most attempts at unpacking heterogeneity in strategic reorientations by incumbent firms have focused on organizational-level explanations (Cockburn et al. 2000, Hill and Rothaermel 2003, Tripsas 1997). Ultimately, however, any response to technical change is constructed through the decisions made by top managers within the organization. Research has shown that executives play an important role in directing various types of strategic renewal, including organizational reorientations (Tushman and Rosenkopf 1996, Virany et al. 1992), new product launches (Boeker 1997, Ciborra 1996, Eisenhardt and Tabrizi 1995, Song and Montoya-Weiss 1998), and changes in research and development (R&D) investment strategies (Kor 2006). The managerial cognition literature has suggested that top managers' cognition is the mechanism driving these effects (for a summary see Huff and Huff 2000). The underlying model is one in which strategic action is shaped by how managers notice and interpret change, and then translate those perspectives into strategic choice (Daft and Weick 1984, Reger and Palmer 1996). Specifically, what top managers pay attention to shapes how they decide and act (Cho and Hambrick 2006, Kaplan et al. 2003, Ocasio 1997).

Technological change provides opportunities for incumbent firms to renew themselves, but these opportunities are generally fraught with uncertainty. Where

information about the direction of change is ambiguous, it is not clear that the CEOs of all firms that stand to gain from the change would be aware of its potential. It is also possible that other factors competing for the CEO's limited attention may distract from the emerging opportunity. Under circumstances where the change would require substantial investments with risky payoffs, an understanding of CEO attention may help explain which firms would take advantage of this opportunity and when they would do so (Barr 1998).

Although managerial cognition research usefully identifies the importance of cognitive effects, sorting out the difference from organizational-level effects is difficult because these alternative explanations have not typically been included in analyses. Recent studies have made the case that it would be important to consider the effects of managerial and organizational effects simultaneously. King and Tucci (2002) and Kaplan (2008) have argued that managerial factors moderate organizational-level effects on a firm's pace of adaptation to technical change. McGrath (2001) found that managerial capabilities affect organization-level exploration and learning. Tripsas and Gavetti (2000) showed that managerial beliefs and organizational capabilities must be aligned for strategic change to occur. From this work, we conclude that managerial effects on outcomes will be separate from those of organizational capabilities. That is to say, managers are not simply making unproblematic matches between the organization and the environment. These studies also show that managerial cognition and organizational orientations are often highly interrelated and that it would be difficult to understand firm action in the face of change without taking into account their interactions.

To explore these possibilities, this paper examines the relative and interrelated effects of managerial and organizational factors on one specific type of strategic renewal—entry into a new product market by industry incumbents. We contrast measures of CEO attention with those of organization-level orientations. Because attention is the “noticing, encoding, interpreting, and focusing of time and effort by organizational decision-makers” (Ocasio 1997, p. 189), it captures the presently evolving focus of a CEO's cognitive effort. An organizational orientation toward a particular issue is the best organization-level corollary for attention because it captures the importance of that issue to the firm (Kohli and Jaworski 1990, Narver and Slater 1990).¹ The greater a firm's orientation toward a specific technology or industry, the more likely the firm will have developed related capabilities, processes, complementary assets, and incentives (Barney 1991, Nerkar and Roberts 2004). The concept of organizational orientation recognizes that investing in capabilities and complementary assets will have attendant effects on organizational incentives because firms have an interest in using and protect-

ing these investments (Caves 1984, Mitchell 1989). We will look at the expected independent effects of organizational orientations and CEO attention in each of the three directions outlined earlier—toward the emerging technology, the existing technology, and the affected industry—and also consider the expected interactions between attention and orientation.

2.1. Effect of Direction of CEO Attention on Timing of Entry

In this section, we first explore the effects on strategic renewal of CEO attention and organizational orientations toward technical issues and then turn to industry-related factors. Strength in technical knowledge has been shown to help a firm recognize new information, assimilate it into the organization, and translate it into a new product in the face of radical technical change. This has been called absorptive capacity by Cohen and Levinthal (1990), who show that higher levels of general technology knowledge, as measured by R&D spending, create greater abilities to adapt to new technologies (see Zahra and George 2002 for a review). Investments in technical knowledge at the organizational or managerial level can improve access to technological information (Gatignon and Xuereb 1997) and may be the “ticket of admission to an information network” (Rosenberg 1990, p. 170), providing both better guidance for applied research and the ability to access and apply scientific knowledge developed elsewhere (Gambardella 1992). However, such basic technical capabilities have been shown to exert inertial forces on incumbent firms because they become core rigidities (Leonard-Barton 1992) or competency traps (Levitt and March 1988). These different perspectives on the impact of technical knowledge are likely based in large part on differences in the types of knowledge being developed and accumulated. It may be more instructive to examine those program-specific investments in technical capabilities that might favor adaptation and others that might be associated with inertia (Cockburn and Henderson 1998, Cockburn et al. 2000). The technical orientations we focus on in this paper relate to the emerging and existing technologies, and we argue that the former will have an accelerating effect on the timing of entry into a new product market and the latter a dampening effect.

Clearly, technical knowledge should accelerate entry if it is focused on the emerging technology. At the organizational level, firms that have accumulated technical experience useful in competing in a new market would be much more likely to pursue the emerging opportunity quickly (Helfat 1997). The literature on related diversification suggests that firms are more likely to enter closely related market spaces when they already possess the relevant organizational capabilities to be competitive in the new market (Chatterjee and Wernerfelt 1991). More importantly for our research, Cattani (2005) shows that communications firms with “preadaptive” organizational

knowledge related to optics (resident knowledge within the firm that was specifically applicable to the new technological paradigm) accelerated research efforts into the new technology. Because the evidence in support of the relationship between organizational orientation toward the emerging technology and strategic action is strong, we do not hypothesize about it here. However, we control for this underlying organizational orientation before examining other orientations.

Research has already shown that CEO attention to an emerging technology is associated with the subsequent investment in the development of such technical capabilities (Kaplan 2008, Kaplan et al. 2003). A primary role of managers is to identify new opportunities for the firm (Augier and Teece 2009) and insert themselves to adapt the microprocesses that enable strategic renewal (Salvato 2009). Attention to a new technology implies that a CEO has already overcome significant barriers to accepting its importance. These barriers may be derived from structural or cognitive information processing constraints (Nelson and Winter 1982, Williams and Mitchell 2004) or ineffective or local search routines (Gavetti and Levinthal 2000, Katila and Ahuja 2002). The question is whether CEO attention might have an effect further down the product pipeline on the timing of entry into the product market beyond the direct effect on capability development. Because product innovation in a new technical market may be seen as a risky activity within an incumbent firm, it is the role of top managers to legitimize and support the new technology to ensure its adoption by the organization as a whole and not just in the R&D department (Dougherty and Heller 1994). CEO attention to the new technology would be a signal for the organization to speed this risky action.

HYPOTHESIS 1 (H1). *Greater CEO attention toward the emerging technology will be associated with earlier entry into a new product market based on that technology even when controlling for the related organizational orientation.*

Whereas a focus on an emerging technology may accelerate entry, many theories predict that an orientation toward *existing* technologies could have the opposite effect. One mechanism by which this retarding effect could take place at the organizational level is through routinization and organizational inertia. Firms could be wedded to technologies that become obsolete (Christensen et al. 1998). Constrained by local search related to existing routines, firms may not be able to move to radically new technologies (Martin and Mitchell 1998, Rosenkopf and Nerkar 2001). Entry therefore could be severely delayed or derailed because of the mismatch between the new market needs and existing capabilities (Miller and Chen 1994, Teece et al. 1997). Firms with a deeply embedded orientation toward the existing

technology will find it difficult to substitute new knowledge in for the old, even if the old knowledge becomes obsolete (Lavie 2006). In the case where the impending radical change requires significant but highly uncertain investment in the new technology, firms that are strongest in the existing technology may rely on their current competencies until the uncertainty is reduced, thus delaying adoption (Anderson and Tushman 1990). Similarly, an incumbent may delay entry to protect the rents derived from existing investments in capabilities (Gilbert and Newberry 1982, Henderson 1993, Reinganum 1983). The implication is that if such existing knowledge is not directly useful in a new market, it will not speed up and may even hinder efforts to launch a product into the market.

HYPOTHESIS 2A (H2A). *Greater organizational orientation toward the existing technology will be associated with later entry into a new product market based on an emerging technology.*

At the CEO level, attention to the specifics of the existing technology may indicate an overinvestment in the existing technological trajectory of the firm (Christensen et al. 1998, Dosi 1982). Even though part of the manager's task is to navigate search processes and set a course for the organization, a manager with a strong focus on the firm's existing technological processes may be unable (or unwilling) to support the necessary distant search that would help the firm adopt radical technical change (Levinthal 1997). CEO attention to the existing technology signals that the firm's identity is linked to the specific technological platform, which impedes the ability to adapt (Tripsas 2009). Managers might get stuck in an existing mindset and therefore prevent the firm from entering a new product market even if the organization has the salient capabilities (Tripsas and Gavetti 2000). These managers may be so immersed in the technical details of the firm's existing business that they are unable to see (or unwilling to accept) technological changes on the horizon or the necessary effort that would be required to take that technology to market.

HYPOTHESIS 2B (H2B). *Greater CEO attention to the existing technology will be associated with later entry into a new product market based on an emerging technology even when controlling for the related organizational orientation.*

Industry experience has also been shown to shape organizational response to new technologies. At the organizational level, the economic argument is that incumbents will enter new product markets in industries where they already have deep experience to create user switching costs (Lieberman and Montgomery 1988) or to preempt competition (MacMillan 1983). If a firm's revenue portfolio is dominated by the industry where the new technology will be most important,

then the technology should be recognized and adopted more quickly (Cockburn et al. 2000). The motivation for entry is therefore to protect an existing position or investments in complementary assets (Mitchell 1989). Industry experience also allows for the long-term development of industry-specific capabilities and complementary assets that shape competitive advantage (Dierickx and Cool 1989, Nerkar and Roberts 2004). Complementary assets such as industry connections, know-how for serving a customer, relationships, and other industry knowledge have been shown to improve incumbent firm performance in new product markets (Teece 1986). This is particularly true if the new technology will be used by existing customers, as in the case of typesetting (Tripsas 1997). Similarly, Klepper and Simons (2000) have shown that firms experienced in the radio market did better in the emerging television market due to the transferability of certain industry-specific capabilities from one setting to another. Thus, we expect that an organization's orientation to the affected industry would help it manage the launch of a product in this new arena to the extent that existing customers can make use of the new technology (as is the case in our study).

HYPOTHESIS 3A (H3A). *Greater organizational orientation toward the affected industry will be associated with earlier entry into a new product market based on an emerging technology.*

Less is known about how industry-related experience plays out at the CEO level. We suggest that top managers with substantial experience in an industry would be more aware of competitive actions (Greve 1998, Johnson and Hoopes 2003), thus directing the firm to adopt technologies that would appeal to their customers. Their dependence on the existing value network would increase the likelihood that they would respond to these pressures and incur the costs and risks to launch a new technology into the market.²

HYPOTHESIS 3B (H3B). *Greater CEO attention to the affected industry will be associated with earlier entry into a new product market based on an emerging technology even when controlling for the related organizational orientation.*

2.2. Relationship Between CEO Attention and Organizational Orientations

The arguments up to this point have attempted to highlight the separate effects of CEO- and organizational-level orientations. It seems likely, however, that these effects are also interrelated. Adner and Helfat (2003) proposed that dynamic managerial capabilities enable managers to reconfigure organizational competencies in the face of changes in the industry. In times of radical technical change, it makes sense that managerial interpretations about the nature and importance of the change

may have an impact on how the firm's capabilities and resources are mobilized. But the effect of such dynamic *managerial* capabilities may in part depend on the existence of organizational capabilities (King and Tucci 2002). Could the delaying effect of investment in capabilities related to the existing technology be reversed or mitigated by increased managerial attention to the emerging technology? This would suggest the possibility of "compensatory fit" (Gulati and Puranam 2009), where one level of the organization (CEO attention) compensates for impediments at another level (technological orientation). Conversely could the accelerative effect of an organization's orientation to the industry be amplified by increased managerial attention to the emerging technology? We explore the extent to which the value of CEO attention to the emerging technology is contingent upon the firm's possession of certain capabilities through an examination of the interactions between these factors.

Despite the expected inertial impact of an organizational orientation toward the existing technology discussed earlier, this sustained orientation does represent an accumulation of technical capabilities, albeit in the existing technologies. To the extent that these capabilities may be transferable to the new regime (Breschi et al. 2003, Griliches 1992, Hur and Watanabe 2001), it is possible that they may actually provide motivation for the firm to enter the new market, as long as the CEO is aware of the potential to refocus these existing capabilities to the new market context. However, this may only occur in the case of significant levels of CEO attention toward the emerging technology, which could reverse the inertial effect of these capabilities. In this situation, the CEO is presumably creating motivation to reconfigure capabilities for the emerging market. The validity of this hypothesis obviously depends upon the potential of the specific technical capabilities to be adapted or the existence of general technical capabilities that might be of value in multiple market contexts.

HYPOTHESIS 4 (H4). *CEO attention to the emerging technology will mitigate the delaying effect on the timing of entry of the organizational orientation toward the existing technology.*

Although an organizational orientation toward the industry may enable a firm to enter with a new technology more quickly, this process might simply build up a stock of complementary assets that may or may not be used to facilitate transition into a new product market (Nerkar and Roberts 2004). Under these circumstances, the complementary assets, although valuable, may not be focused on the proper targets if senior managers are not open to their application to a new technological opportunity. If capacity constraints to applying these capabilities exist, managers will be forced to make allocation decisions that affect their usage (Levinthal and Wu 2008).

Where the firm has a sufficient stock of experience-based industry assets and capabilities, how top managers view those assets and choose to deploy them may shape the timing of entry (Burgelman 2002). Effectively, CEO attention may only be useful in speeding transition when there are industry-related assets to be directed toward the new and emerging technological opportunity.

HYPOTHESIS 5 (H5). *CEO attention to the emerging technology will amplify the accelerating effect on the timing of entry of the organizational orientation toward the targeted industry.*

3. Methods

3.1. Research Setting and Sample Construction

This study examines the factors shaping the decision by incumbent communications technology firms to develop and commercialize an optical product during the fiber-optic revolution. This radical technical change shook up the industry landscape over the course of a decade. The focal actors in this study are the incumbent firms who supply the hardware, including cables, routers, switches, and other devices, to build the communications infrastructure. This setting provides an especially appropriate test of the hypotheses we explore in this study because optics represented a potentially substantial opportunity to these firms, but adoption involved high risks and uncertainties.

CEOs in the communications technology industry would not have taken the decision to enter the fiber-optics market lightly. The choice of when and how to take advantage of the creation of fiber-optic networks represented one of the most important strategic decisions incumbents in this industry faced. On the one hand, incentives to adopt optics were high due to the attractiveness of the technology to a critical set of communication technology firms' customers, the telecommunications service providers (carriers). Fiber optics would drop by orders of magnitude the cost of voice and data transmission for these carriers (Bray and Baney 1997). Thus, optical technologies were not "disruptive" (Christensen and Bower 1996) in that they were useful to existing customers. Firms that entered this product market were not exploiting new niches, but rather were addressing existing needs with a new technology. Incumbents would stand to benefit more than *de alio* or *de novo* entrants from the introduction of this kind of radical technology because they would likely possess the requisite complementary assets such as distribution channels and sales relationships to make the launch successful.

On the other hand, despite the potential attractiveness of the opportunity, there was still a great deal of risk and uncertainty present. For incumbent firms, fiber optics represented a radical innovation (Henderson 1993) that was competence destroying (Tushman and Anderson 1986)

in that entirely new technical expertise was required (Fine and Kimerling 1997). Thus, the change studied in this paper is not the kind of incremental improvement that all incumbent firms would logically be expected to adopt quickly and efficiently. Outcomes were uncertain because, especially in the early days, it was not clear which specific technologies should be pursued nor whether the complex equipment would work. Because investment costs to enter the fiber-optic product market were extremely high, the technological uncertainties presented the incumbent firms with a risky choice.

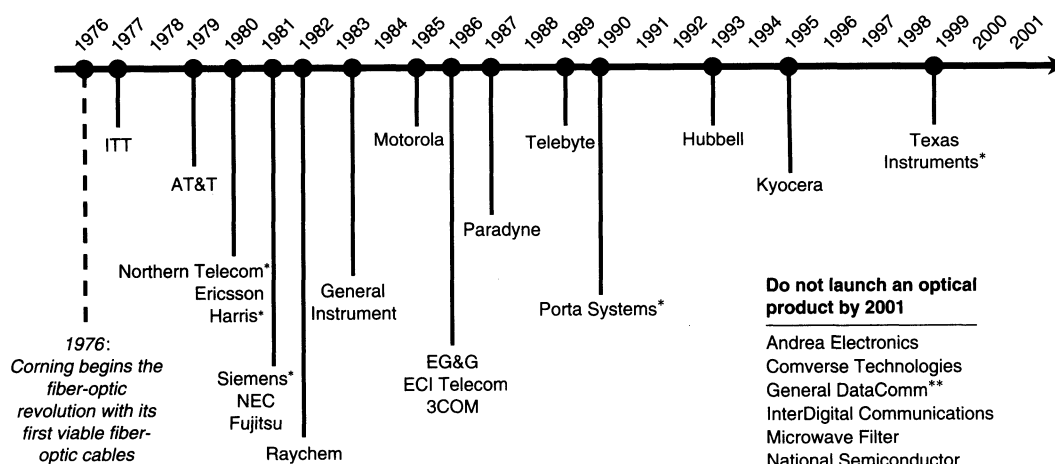
Our initial sample contained 29 publicly traded communications technology firms that were incumbents in 1985 at the time of the first viable optical communications system.³ Twenty-one were incumbents in 1976 at the advent of the first fiber-optic cables and eight entered the industry soon afterward (but with nonoptical products). The sample does not include Corning, who started the optical revolution in 1976 as a *de alio* entrant. This group represents a stratified sample of incumbent telecommunications hardware firms on U.S. stock exchanges. Large incumbent firms are oversampled—all firms are included in the sample (19 firms). The medium (five firms) and small firms (five firms) in the sample are randomly chosen from the respective firm populations.⁴ Three firms were dropped for insufficient data (two large firms and one small) for a final sample of 26 firms. The study covers optical product market entry activity from 1976 to 2001. In this period, 20 of the 26 firms entered, and 6 did not. Only one of those six firms is right-censored due to organizational failure before entry into the fiber-optics market (General Datacomm), and this failure occurred in 2000—one year before the end date of the study. There are 307 firm-year observations, and the median number of years to entry per firm is 12.

3.2. Variable Construction

3.2.1. Dependent Variable. Consistent with the product introduction literature (Mitchell 1989), we have focused our attention on the launch timing of the firm's first commercial optical product. The dependent variable is a dummy based on whether or not the firm introduced a product specifically for the fiber-optics product market in that year—either through internal development (15 of the 20 cases) or external activities (acquisition, joint ventures, or licensing, which were 5 of the 20 cases).⁵ Figure 1 provides a timeline for entry indicating type of entry mode for firms in our data set. Information on entry into the fiber-optics market was obtained from company press releases, trade journal articles, and other available industry data sources. This triangulation across multiple sources of data allowed us to ascertain that a product was actually launched rather than simply announced.

3.2.2. Organizational Orientations and CEO Attention: Key Independent Variables. We measure the focus on the emerging technology, the existing technologies,

Figure 1 Timing of Entry into Fiber-Optic Product Market by Incumbent Communications Technology Firms



*Firms entering through acquisition, joint venture, or licensing.

**Organizational failure in 2000.

and industry in terms of both organizational orientations and CEO attention. In the context of this study, *emerging technology* focuses on optical and optics-related technologies, *existing technology* focuses on nonoptical communications technologies in use in 1976, and *industry* focuses on factors related to the telecommunications industry. Organization-level measurements are taken from financial and patent data. Our measures of CEO attention are derived from the letters to shareholders in the company annual reports. Each of these measures is discussed in detail below. Because entry into a new product market is realized some time after the decision is made, we calculate our orientation and attention variables as lagged stocks to capture the prevailing circumstances within the firm at the time of the decision to enter the new market. Stock variables allow for the dominant impact of one year's activity as well as for recent trends, and are therefore preferred to a noisier flow measure. We used a three-year decay period. In addition, all variables are lagged one year. As an example, a market entry in 1980 would be predicted by the stocks of organizational orientations and CEO attention from 1977 to 1979.

Organizational Orientations. Although investments in organizational technical orientation have typically been measured using R&D spending, this information is only available from firms at an aggregate level. To get at the program-specific level of investment in capabilities required for this study, we use patent data. Because patents are categorized by technology classifications, this allows us to break out orientations to the emerging optical technologies from those associated with the existing nonoptical communications technologies. Although patents, especially as weighted by citations, have often been used to measure innovative output (Jaffe and Trajtenberg 2002), they may reasonably represent innovative input as a proxy for the degree of investment in a particular technology (Schmookler 1972). Patents can be construed to be an indicator of the direction of R&D

in the firm (Henderson and Cockburn 1994). Qualitative evidence from interviews in the industry suggests that this is particularly true in the communications technology industry, where patenting is assumed to be a natural way to document research efforts. One might be concerned that patenting only represents successful R&D activities, and therefore does not capture failed investments in research. For the purposes of our study, the potential bias this introduces only makes such measures of technology investment a conservative test of their effects on outcomes.

To understand the orientation to the emerging technology at the organizational level, we measure optical patenting activity, categorizing those patents that apply specifically to optical technologies as a percentage of all patents (*optical patents/patents*).⁶ This is used as a control in all of our regressions. To understand the orientation toward existing technologies, we measure the percentage of nonoptical communications-related patents the firm has in its portfolio (*commtech patents/patents*).⁷ Both of these patent-based variables are measured at the date of application. We measure orientation toward the telecom industry through the percentage of the company's overall sales that were reported as coming from the general telecommunications sector (*telecom sales/sales*). As mentioned above, the notion of orientation captures the accumulation of investments in capabilities and complementary assets in a particular arena. Therefore, the higher the ratio of telecom sales, the more focused and dependent the firm is on the telecommunications industry and the greater the likelihood the firm has developed industry-specific competencies or complementary assets. This would be indicative of the balance of power and the alignment of organizational incentives toward serving that industry (Cockburn et al. 2000).

CEO Attention. Finding an appropriate measure of CEO attention was an important empirical challenge for this study (Lant and Shapira 2001). Some studies have

used managerial demographics as a proxy for cognition (Norburn and Birley 1988, Virany and Tushman 1986), but these are relatively fixed. Given the rapidly changing nature of the market and the fact that attention can shift over time (Hambrick et al. 1993), we sought a measure that would capture more directly time-varying attention to issues. The most reasonable place to look for information on CEO attention would be statements made by CEOs, but many typical sources would not be appropriate for a study that requires consistent information across firms and over time. Such statements cannot be captured through surveys or interviews because of the lack of availability of informants over long periods of time and the inherent risks of retrospective bias. CEOs' comments in speeches, media interviews, or conference calls with analysts are ad hoc and therefore not available in comparable forms for all firms in all time periods.

An increasingly frequently used alternative source for measuring attention is the letter to shareholders in the annual report (Barr et al. 1992, Cho and Hambrick 2006, Kaplan 2008, Kaplan et al. 2003, Osborne et al. 2001). To assess the emphasis on various themes in these statements, we use word counts derived from these documents. Content analysis techniques such as word counts have been acknowledged as a valid way of capturing attention (Duriau et al. 2007). We normalize the counts by the total number of words in the letter to produce a measure of the attention relative to other themes. This measure of frequency is a proxy for importance to managers (Abrahamson and Park 1994, Cho and Hambrick 2006, Wade et al. 1997). Such a thematic approach to textual analysis is the most appropriate given the need to generate quantitative measures over a large number of texts (Roberts 1997). The appeal of this measure is that it is both replicable and prospective (Bettman and Weitz 1983, Kiesler and Sproull 1982).

Though some construe letters to shareholders solely as outcomes of symbolic management rather than as representative of leaders' real actions (Westphal and Zajac 1998), there are theoretical and empirical reasons that we believe these documents are important evidence of CEO attention. First, though much of the annual report is likely developed by the public relations department in a firm, evidence suggests that the letter itself is written or closely edited by the CEO (Cho and Hambrick 2006, Kohut and Segars 1992). Second, Fiol's (1995) study comparing internal statements in strategic planning documents and external statements in annual reports showed that the basic themes were the same, even if positive or negative interpretations were not significantly related in the two kinds of documents. Finally, from the data in our research, we see clear signs that changes in CEOs (though relatively rare) had a dramatic impact on the style, length, and content of the letters. For example, the majority of the 20 CEO changes covered in our sample

were associated with an at least 20% change in the average number of words used in the letter. Such changes are signals that each CEO put his or her imprint on the letter to shareholders. We also noticed significant shifts in tone and approach used by new CEOs, even if the CEO had been a member of the senior leadership team for quite some time prior to assuming the top job.

One might be concerned about the possibility that CEOs may not communicate their actual views to their shareholders, either because they are keeping the information secret or because they are dissembling to gain strategic advantage over rivals (Crawford 2003). We believe that an excessive amount of misdirection by the CEO is unlikely because the letter to shareholders goes out under the CEO's signature, and therefore, for fiduciary reasons, it is unlikely that the CEO would suppress important discussions (Abrahamson and Park 1994). Also, to the extent that there is a gap between reported and actual attention to issues, this potential bias would simply make our test of the association between attention and outcomes more conservative. Finally, because we were careful to ensure that our dependent variable is based on actual product launches, not just announcements of launches, there is no common source bias in the data.

Of the 370 potential letters in our data set, we have located 325 letters. Another 45 letters could not be obtained from the companies, any library, or the SEC and were likely not published. Nearly all of the missing documents came from the years 1973–1976, when the values are a small part of the decaying stock variables we construct. We interpolated missing data, and robustness checks excluding interpolated data points did not produce any significant changes in the results. Our analysis included three different word counts: the words were selected by reviewing a concurrence of the more than 26,000 individual words that appear in the 325 letters. All relevant words for each orientation were selected (see the appendix for a list of the words used). To measure attention to the emerging technology (*normalized optical words*), we selected all words associated with optics in general (such as fiber optics) as well as with specific optical technologies (e.g., lasers or erbium doped fiber amplifiers). To measure attention to the existing communication technologies (*normalized commtech words*), we selected all words mentioning a specific communications technology other than those associated with optics, such as handset, amplifier, or semiconductor. To measure attention to the industry (*normalized telecom words*), we selected all words related to the industry in general and its main business (e.g., telephony, communications, or WAN—wide area network). We normalized each by dividing by the total number of words in each letter. To reduce the size of the reported coefficients (due to the small size of these variables), each CEO attention variable is multiplied by 1,000 for the regressions. As with the organizational orientation variables discussed earlier,

we calculate these attention variables as lagged, three-year decaying stock variables to capture attention in a window of time.

3.2.3. Control Variables. In addition to our control for prior related competence (organizational orientation to optics as measured by optical patents as a percentage of total patents), we selected several other controls to account for alternate explanations as to why some incumbents might enter the fiber-optics product market more quickly than others. Because available firm resources may effect entry abilities (Cheng and Kesner 1997, Singh 1986), and there is evidence of economies of scale in the adoption of new technologies (Agarwal and Audretsch 2001, Zucker and Darby 1997), we control for total sales (*sales*), calculated as a three-year stock variable and logged. We also include a measure of general technical capabilities or absorptive capacity, measured as by Cohen and Levinthal (1990)—R&D as a percentage of sales. Because prior research has suggested that older organizations may have more available resources to pursue new opportunities (Hannan and Freeman 1989), and other studies on market entry have discussed relationships between age and the decision to enter new markets (Durand and Coeurderoy 2001, Mitchell 1989), we also include a control for organizational age in years (*firm age*). The final control variable (*smaller*) is a dummy that notes whether the firm was one of the 19 large firms or part of the sample of medium and small firms. Smaller firms might be pursuing a niche strategy to the point that their baseline hazard rate would not be the same as that for the larger firms that competed more broadly in the telecommunications industry. We enter this as a stratification variable.

3.3. Descriptive Statistics and Correlations

Because the foci on emerging technologies, existing technologies, and the industry are measured at both the organizational and CEO levels, we had significant concerns about high levels of intercorrelation. Table 1 shows that the only such problem between CEO attention and organizational orientation is that in the domain of industry focus, where the correlation is 0.590. These are also relatively highly correlated with other variables, raising additional multicollinearity concerns. Their variance inflation factors are also somewhat high, suggesting that we may have problems estimating our models. As reported in the Results section, the high level of multicollinearity with these two industry variables does indeed make it difficult to tease apart the differences in effect between the organizational and CEO levels in this area. There are also particularly high correlations and variance inflation factors between the *smaller dummy* and other controls. This suggests that it is more appropriate to use this variable as a stratification variable, affecting the baseline hazard in any given year but not affecting the estimation of the entire model directly. We will revisit multicollinearity in the discussion of the results from regressions below.⁸

Given the findings of prior work on the relationship between CEO attention to optics and subsequent optical patenting behavior (Kaplan 2008), there are potential concerns about endogeneity when we use both of these variables to predict subsequent optical product introduction. There are two specific empirical reasons that we believe this endogeneity is not a significant concern in this study. First, Kaplan (2008) has demonstrated a three-year lag between CEO attention to optics and the filing of optical patents. Given our consideration of these two variables as contemporaneous and as three year decaying stocks, there is little overlap between the

Table 1 Correlations and Descriptive Statistics ($n = 307$)

	1	2	3	4	5	6	7	8	9	10	11
1 <i>Launch of optical product</i>	1.000										
2 <i>Normalized optical words</i> ^a	0.081	1.000									
3 <i>Commtech patents/patents</i> ^a	−0.043	0.128*	1.000								
4 <i>Normalized commtech words</i> ^a	−0.045	−0.105	−0.021	1.000							
5 <i>Telecom sales/sales</i> ^a	−0.024	0.107	0.406*	−0.303*	1.000						
6 <i>Normalized telecom words</i> ^a	−0.027	−0.057	0.576*	−0.109	0.590*	1.000					
7 <i>Optical patents/patents</i> ^a	0.208*	−0.042	−0.123*	−0.039	−0.166*	−0.047	1.000				
8 <i>Sales</i> ^a	0.160*	−0.141*	−0.140*	0.345*	−0.551*	−0.411*	0.485*	1.000			
9 <i>R&D/sales</i> ^a	−0.075	−0.116*	0.332*	0.030	0.192*	0.569*	−0.153*	−0.445*	1.000		
10 <i>Firm age</i>	0.151*	−0.231*	−0.231*	−0.027	−0.213*	−0.345*	0.474*	0.466*	−0.261*	1.000	
11 <i>Smaller dummy</i>	−0.136*	0.183*	0.407*	−0.263*	0.794*	0.527*	−0.443*	−0.758*	0.308*	−0.506*	1.000
Mean ^b	0.065	0.001	0.212	0.002	0.652	0.006	0.011	5.789	0.105	41.570	0.456
Standard deviation	0.247	0.001	0.302	0.003	0.383	0.006	0.022	2.986	0.161	29.128	0.499
Minimum ^b	0.000	0.000	0.000	0.000	0.000	0.00	0.000	0.000	0.007	3.000	0.000
Maximum ^b	1.000	0.007	1.000	0.022	1.000	0.033	0.108	11.242	1.000	135.000	1.000
VIF (variance inflation factor)		1.23	1.81	1.28	4.04	3.35	1.71	4.04	2.16	1.95	6.29

^aCalculated as a three-year decaying stock variable.

^bUncentered and unlogged means, minimums, and maximums reported.

* $p < 0.05$.

measures in ways that would cause endogeneity. Second, it is worth noting the low correlation between the optical patenting and optical attention variables (-0.08). The lower alignment between organizational orientation and CEO attention is likely due to the rapidly evolving nature of the emerging optical technologies in this early phase of technology development. Because the CEO is usually quite removed from the individual members of an R&D organization, differences in focus throughout the organization are likely (Gavetti 2005). In a sample of firms that have not yet actually adopted optical technologies, it may not be surprising that R&D efforts and CEO attention to these issues do not correlate highly. These factors suggest that any endogeneity problems will not be too severe. We verify this with a test of the mediation of attention by the patenting variable (Baron and Kenny 1986). We find that the effect of cognition is biased downwards when we control for patenting, due to partial mediation, and conversely, the effect of patents is biased upwards when we do not control for cognition. When controlling for cognition, though, the effect of patents should not be biased. As a result, we simply control for optical patenting in all regressions and look at the effects of attention to optics above and beyond those absorbed in the orientation measure.

3.4. Data Analysis Procedures

We use event history methodology to model the likelihood of first introducing an optical product for each firm in the risk set in any given year. Because we are interested only in the first introduction of a product (a nonrepeating event) and test time-varying covariates, we utilize a Cox proportional hazard model to understand the effects of different factors on the process.⁹ Given the coarse measures of the dependent variable, an alternative approach would be to use a discrete time model. The results are robust to this alternative specification. We used the Efron method to account for ties (when two or more firms enter the market in the same year) because this is a more robust means of handling ties than the simpler Breslow method. Given that there were a modest number of ties in the data, the method that we chose to handle them did not have any significant impact on the performance of the model in various robustness checks. To avoid collinearity problems for the main effects in the tests of interactions, we use mean-centered variables for all regressions (though we report uncentered means, minimums, and maximums in Table 1 for informative purposes). We focus our analysis on the full model with all variables included. However, given the relatively small sample size (307 observations of 26 firms and 20 event occurrences) and high levels of intercorrelation, we were concerned that the full model might overly stress the data. Therefore, we also include intermediate models evaluating each hypothesis independently. The results for our variables of interest are generally consistent across the models.

4. Results

In this section, we evaluate the evidence for each of our hypotheses, first comparing CEO and organizational effects and then examining the results for interactions.

4.1. Effect of Attention and Direction of Attention on Timing of Entry

The results of the event history model are reported in Table 2. Looking at the control variables in the full model (Model 6) and intermediate models, *optical patents/patents* performs as expected. Developing a competency in the new technological regime is an important input to launching relevant new products. Additionally, the coefficient on *sales* is positive in intermediate models and positive and significant in the full model, indicating that larger firms are more likely to have the requisite resources to address new market opportunities and that there may be economies of scale in doing so. The coefficient on *R&D/sales* is negative and significant in the full model. Increased R&D spending after controlling for patenting in the emerging technology is associated with slower launch of new optical products, because firms may have too many other active research trajectories. The coefficient on *firm age* is not significant in any model.

We first address Hypothesis 1, regarding the direct relationship between CEO attention to the emerging technology and product launch. As discussed earlier, prior work has already shown a relationship between CEO attention to the emerging technology and R&D investment in that technology (Kaplan 2008). The model presented here looks for a direct relationship between attention and product launch, beyond the indirect relationship through patenting (already controlled for in the model). The full model (Model 6) suggests that a weak direct relationship may in fact exist, and thus the effect of attention to optics is not fully mediated by optical patenting. Incremental CEO attention to optical technologies up to four years before the initial optical product launch is associated with differences in actions among incumbent firms. This finding, as well as the stronger signs of significance in most of the intermediate models, provides weak support for H1.

We do not believe that this finding is an indication that CEO attention to optics (at least as measured using letters to shareholders) is simply a symptom of a rising tide in the industry or the firm. One might be concerned that a discussion of optics in the letter would just be acknowledging ongoing R&D activity and imminent launch of a new product, but our evidence suggests that this is not the case. First, the extremely low correlation (-0.082) between *normalized optical words* and *optical patents/patents* is a sign that these two variables are measuring different things. Second, in regressions not reported here, we also used optical patenting activity to predict subsequent CEO attention to optics. Finding such a relationship would show that patenting activity was more likely to influence cognition. However, in our

Table 2 Conditional Effect of CEO Attention on Product Launch ($n = 307$): Cox Hazard Model on New Optical Product Launch Date, Stratified on Smaller Variable

	Model 0	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Emerging technology							
<i>Normalized optical words</i> ^a		0.583* (0.250)			0.539* (0.244)	0.373 (0.277)	0.685^ (0.393)
Existing technology							
<i>Commtech patents/patents</i> ^a			2.738* (1.324)		2.784* (1.234)		3.728** (1.356)
<i>Normalized commtech words</i> ^a			−0.300** (0.088)		−0.306** (0.094)		−0.577** (0.196)
Industry							
<i>Telecom sales/sales</i> ^a				0.756 (1.205)		0.920 (1.092)	0.473 (0.881)
<i>Normalized telecom words</i> ^a				0.132^ (0.071)		0.184* (0.093)	0.286* (0.132)
Orientation × Attention interaction							
<i>Commtech patents × Optical words</i>					−0.160 (0.812)		−0.160 (1.421)
<i>Telecom sales × Optical words</i>						2.371* (0.978)	3.965*** (1.133)
Controls							
<i>Optical patents/patents</i> ^a	27.637*** (6.717)	25.507*** (7.256)	36.553*** (10.057)	12.217 (11.886)	33.750** (10.681)	12.356 (12.102)	26.055* (11.800)
<i>Sales</i> ^a	0.153 (0.181)	0.147 (0.211)	0.073 (0.141)	0.275^ (0.155)	0.067 (0.168)	0.401^ (0.227)	0.575* (0.239)
<i>R&D/sales</i> ^a	−9.519 (6.654)	−5.763 (6.610)	−9.175 (6.185)	−12.483^ (6.836)	−5.586 (4.512)	−19.265^ (10.502)	−25.904* (11.434)
<i>Firm age</i>	0.006 (0.006)	0.012 (0.007)	0.001 (0.001)	0.010 (0.009)	0.006 (0.008)	0.011 (0.011)	0.001 (0.010)
Log likelihood	(33.5)	(31.2)	(29.7)	(31.2)	(28.0)	(27.2)	(22.1)

Notes. All independent variables lagged one year.

^aCalculated as a three-year decaying stock variable.

^ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

results, the coefficient on optical patenting is not significant. We therefore conclude that our measure of CEO attention to optics is not simply an epiphenomenon of ongoing organizational activities.

Hypotheses 2A and 2B address organizational orientation and CEO attention toward the existing technologies within the industry, which we argue should delay entry into optics. At the organizational orientation level, we find evidence directly counter to H2A. Greater orientation toward the existing technology (*commtech patents/patents*) is associated with more rapid introduction of a product based on the emerging technology (positive and significant at the 0.01 level in the full model). The results for CEO attention (*normalized commtech words*) provide support for H2B because the coefficient is negative and significant at 0.01 in all models. CEO attention to existing technologies is associated with delays in the timing of entry. The combination of the findings at the organizational and CEO levels suggests that, at least in this industry setting, the translation of technical know-how from the application of one technology to another may be relatively straightforward, and that firms may seek to take advantage of those capabilities. Although optical

technologies are radically new in some ways, knowledge of the existing communications infrastructure that comes with an orientation toward existing technologies may still be helpful in making products of use to customers that have systems built from these legacy technologies. The inertia related to a strong focus on the existing solutions attributed in prior literature to rigidity in *organizational* capabilities (i.e. Leonard-Barton 1992, Tripsas and Gavetti 2000) appears instead to be driven by *managerial-level* cognition.

Next, we consider the evidence for Hypotheses 3A and 3B concerning the degree to which organizational orientation and CEO attention toward the telecommunications industry accelerate entry into new optical products. We argue that this will be the case when a radical new technology will be useful to the existing customer base as in this context. Each model, including the full model, shows that the organizational orientation variable is not significant, but that the CEO attention variable is positive and at least marginally significant (at 0.05 in the full model). Greater CEO attention to activities within the specific industry experiencing the technological change is associated with more rapid introduction of a new optical

product in that market (even when controlling for the percentage of the organization's sales that come from that industry). This finding supports H3B, although the high (0.590) correlation between the organizational- and CEO-level variables and the fact that the coefficient on each tested separately (not reported here) is positive and significant suggests that each may have an effect, but that the CEO attention variable is more consistently related to entry timing.

These three findings about the effect of CEO attention on the timing of market entry provide confirmation of the first essential thesis of this research—that the *direction* of CEO attention is an important part of understanding how incumbent firms respond to the opportunities and challenges presented by a radical new technology. The effects of the attention variables on the likelihood of a firm introducing a new optical product are also economically significant. For one standard deviation increases in *normalized optical words* and *normalized telecom words*, the firm is 138% and 490% more likely, respectively, to introduce an optical product in any year. The comparable effect for organizational orientation to the emerging technology is only a 78% increase (the coefficient for industry orientation is not significant). For a one standard deviation increase in the value of *normalized commtech words*, a firm is 87% less likely to enter (the corresponding organizational orientation works in the other direction, producing a 208% increase in likelihood).¹⁰ Although directly comparing coefficients and economic effects is difficult, it is clear that a model ignoring CEO attention would not be as powerful in explaining the heterogeneity of incumbent action as one which takes these cognitive factors into account.

4.2. Relationship Between CEO Attention and Organizational Orientations

Whereas the results above suggest that a CEO's attention to an emerging technology is associated with faster entry in the new product market, we have also argued that such attention may depend on the effects of organizational orientations. To test the hypotheses about the interactions between CEO attention to the emerging technology and organizational orientations, we turn to the interaction terms in Table 2.

Hypothesis 4 suggests that the impact of CEO attention toward the *emerging* technology may depend on the orientation toward the *existing* technology. We find that the interaction term is not significant, and therefore we cannot confirm H4: CEO attention to a new technological opportunity has no significant effect on transforming an organizational orientation toward existing technologies. This may be because an organizational orientation to existing technologies already appears to be positively associated with quicker entry (counter to our initial hypothesis 2A). This effect does not seem to be further accelerated by increased CEO attention to the new technology.

Hypothesis 5 addresses the relationship between an organizational orientation to the industry and the effect of CEO attention to optics. The supposition is that sustained dependence on the specific industry will lead a firm to create potentially useful complementary capabilities as well as the incentives to use them, and that the impact of CEO attention to the emerging technology will be significantly greater under these conditions. This interaction between CEO attention to optical technologies and organizational orientations toward the telecommunications industry is positive and significant (at the 0.001 level in the full model), supporting H5. This result confirms prior findings that managers will choose to enter a new market if the firm possesses the relevant organizational building blocks to enable it to earn higher rents (King and Tucci 2002, Mitchell 1989) and moreover shows that the effect of CEO attention will be amplified when the organization possesses such industry-related complementary assets, capabilities, and incentives. Note that these results run counter to findings in Kaplan (2008) that CEO attention and industry-related organizational incentives are substitutes when driving investments in technical capability building (i.e., patenting). However, this could be explained by the phenomenon of interest here, which is further down the pipeline at the actual point of launch of a product into the market, something that requires the mobilization of industry-related complementary assets. Thus, it is possible that in R&D settings, any kind of impetus pushes research forward (either at the CEO or the organizational level), but once organization-wide efforts are required for product launch, the presence of organizational complementary assets can actually amplify the effect of CEO attention to the emerging technology.

Stepping back, we believe that these results suggest a fairly complex model of organizational action in the face of technological change. It is clearly not the case that the first time a CEO mentioned optics in a letter to shareholders represented the beginning of the firm's association with optical technologies. Instead, it is likely that individual R&D managers with some limited budget discretion would have already allocated resources to all sorts of potentially interesting new technologies, among them optics (McGrath 2001, Thieme et al. 2003). Indeed, 8 of the 26 firms in our sample (including most of the earliest entrants) applied for optical patents at least one year before any mention of optics in the letters to shareholders. Yet, the question is, what motivates a firm to incur the significantly higher risk and expense of moving from R&D investment to launching a new product? The evidence from our analysis suggests that firms may not enter the market until their top managers are sufficiently convinced that the market is viable, attractive, and relevant for the firm. CEO attention to optics and to the industry may be the force that moves a potential new product beyond the research phase and into development and launch. Firms like Nortel and Fujitsu were patenting

in optics in the early 1970s, but did not launch products until 1980 and 1981, respectively, when CEO attention to optical and industry issues rose substantially. However, inertial response may be a result of too much CEO focus on existing technologies.

Therefore, the development of organizational knowledge related to such a radical new technology as optics is not necessarily a complete gauge of when (or even whether) the company will choose to invest the necessary resources to launch that new product. The level of CEO attention is an important additional predictor of the timing of entry, serving to define and legitimize a course of action if focused on the new technology or the industry or to de-emphasize or delegitimize further action if focused on the existing technology. The direction of CEO attention is thus an essential force in shaping the transition from the old to the new.

5. Discussion and Conclusion

In this paper, we set out to explore the conditions under which managerial cognition plays a role in shaping the response by incumbent firms to technical changes affecting their industry. Using a measure of CEO attention derived from company letters to shareholders, we find that managerial cognition is indeed associated with differences in the timing of entry into a new product market. We provide a more contingent view of managerial effects on strategic reorientation which offers one resolution to the debate about whether managers matter. We show that the effect of CEOs differs depending on the focus of their attention—accelerating entry when attention is directed to the emerging technology or the affected industry and slowing response when attention is directed to the existing technology. The effect of attention to the emerging technological opportunity is also dependent on the level of organizational orientation—the capabilities, complementary assets, and attendant incentives—a firm has toward the affected industry. The accelerative effect of CEO attention to the new opportunity matters more when firms possess greater industry-related capabilities. This implies that firms with the same underlying organizational orientations might move in different directions if the CEOs' attention to the situation differs. The effect of cognition is therefore context-specific.

We conclude that both managerial cognition and organizational orientation are important factors in understanding firm response to new technical opportunities, and that considering them simultaneously provides a more nuanced understanding for their relative and interrelated effects. Future research could explore the circumstances under which firm-level orientation and CEO attention would differ from one another. Would such a disparity be unique to dysfunctional (or highly functional) firms, driven by changing industry contexts, a manifestation of

inherent differences between organizations, or the outcome of long-term evolutionary processes? Could attention have a differential effect on reorientation depending on the type of entry mode, potentially having a greater influence when entry occurs by external means such as acquisition or alliance?

The study presented here, like any other, has its limitations. The size of our data set (even if it contains nearly all of the large incumbent firms within the communications technology industry) makes some analyses problematic to perform and interpret. The relatively small number of companies puts constraints on the significance levels of the effects we find. In addition, our choice of context for this research also limits the types of questions these data are suited to address. As pointed out by numerous researchers, radical technological changes can take many forms, and the introduction of fiber optics for communications represents just one change within a single industry. In this setting, the customers for both the old and the new technologies were the same, so we can say less about situations such as those discussed by Christensen and Bower (1996) in which new technologies would be initially targeted to a niche set of customers.

We also focus in this study on strategic renewal by incumbent firms, but it is likely that CEO attention would play an important role in understanding decision-making for both *de alio* entrants from other industries and start-up organizations. Data availability limitations (being unable to measure CEO attention through our method for firms before they start operations), sample identification difficulties (as *de alio* entrants could come from any industry), and the focus on the introduction of the first product (complicating any analysis for firms not already in the market) do not allow us to address these questions here. Other research by one of the authors (Kaplan 2008) confirms the role of CEO attention for start-up firms in shaping technical investments but does not address the later development stage of product introductions. If data were available, it would be instructive to understand how the role of managerial attention may be different or similar for *de alio* entrants and start-up organizations.

Yet, this research has some interesting implications for understanding how firms renew themselves in the face of technical change. Some researchers have attributed the heterogeneity of firm actions to differences in dynamic capabilities that either constrain or enable a firm to put together new resources to address the new product market (Eisenhardt and Martin 2000, Teece et al. 1997, Zollo and Winter 2002). Most attempts to understand dynamic capabilities have operated at the organizational level of analysis (Eisenhardt and Tabrizi 1995, Hill and Rothaermel 2003), but Adner and Helfat (2003, p. 1012) argued recently that they should be understood at the

managerial level as well. Dynamic managerial capabilities, such as managerial cognition, can create or reconfigure organizational capabilities so that the firm can adapt in the face of environmental change. To date, dynamic managerial capabilities have not been measured directly, but instead inferred from organizational outcomes that are assumed to be the product of managers’ decisions (King and Tucci 2002). The analysis presented in this paper fills this gap by measuring managerial cognition (specifically, CEO attention) directly and comparing its effect to that of organizational orientations on firm strategic renewal. Though dynamic capabilities have normally been conceptualized as enabling firms to adapt through multiple periods of change (Teece et al. 1997), our analysis of the ability of managers to navigate their firms through a single, evolving radical change provides some clues to the ways in which managerial cognition could be a source of dynamic capabilities.

By assessing different directions of CEO attention, we can also identify when the effect of cognition might be either myopic or farsighted. Myopia is associated with greater attention to existing technologies and inhibits firm adaptation (Levitt and March 1988). We find that increased organizational capabilities in the existing technologies may not be the source of this myopia, but rather that greater emphasis on existing technologies at the *managerial* level may create cognitive blinders. These blinders make CEOs unable (or unwilling) to consider the possibilities offered by a new technology that lies outside their specific area of expertise. In this line of thinking, top managers who have devoted their attention to improving the firm’s efficiency within the existing technological paradigm may be reluctant to throw the fruits of those

investments away (Anderson and Tushman 1990, Benner and Tushman 2002). However, farsighted attention to a new technology is associated with faster entry. Such attention may communicate priorities to the organization and signal a willingness to make a risky investment to develop a product. In this way, CEOs become powerful champions of resource allocation, incentive construction, and the creation of an organizational climate favoring that technology. Thus, CEO attention can have either myopic or visionary effects on organizations navigating changes in their environments.

In conclusion, this paper demonstrates that managerial cognition is an important factor for understanding heterogeneity in incumbent response to technical change, but that its effect is conditional on both the direction of that cognition and the related organizational orientations. The results suggest that managers do indeed matter: their attention to the industry and technology arbitrate between the organization and its environment, influencing whether and when firms engage in strategic renewal.

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Appendix 1. Words Used to Measure CEO Attention from Letters to Shareholders

Optical words	Commtech words	Telecom words
Color	Amplifier	Communications
Detector	ATM (asynchronous transfer mode)	Data communications
EDFA (erbium doped fiber amplifier)	Attenuator	LAN (local area network)
Fiber optic	Cable modem	Network (not Internet)
Filter	GaAs (gallium arsenide)	Phone
InP (indium phosphide)	Handset	Telecom
Laser	Modulator	Telephony
Light	Router	Traffic
Multiplex or mux	Semiconductor	WAN (wide area network)
Optic	Switch	Wireline
Opto		
Photonic		
SDH (synchronous digital hierarchy)		
SONET (synchronous optical networking)		
Transceiver		
VOA (variable optical attenuator)		
Waveguide		
Wavelength		
WDM (wavelength division multiplexing)		

Endnotes

¹Whereas the notion of organizational orientation was developed in the marketing literature to explore the degree to which the “marketing concept” had been implemented in an organization (including investment in marketing related activities) (Kohli and Jaworski 1990, Narver and Slater 1990), we expand this logic to apply to orientations toward other issues, specifically those related to technology and industry.

²There are circumstances under which a strong industry focus (at the CEO or organizational level) may instead delay entry. Christensen and Bower (1996) show that disk drive manufacturers missed new opportunities because they focused on serving existing customers with different requirements. Experience in a particular niche can lead to deeply embedded ways of doing business that get in the way of entry into new niches that require new practices (Greve 1996). This logic is most applicable when the new technology initially serves *new* customers. In the case where a radical new technology will be useful to the existing customer set (as is the case in our study), we argue that greater industry focus should accelerate entry.

³These data were originally developed as part of the second author’s dissertation (Kaplan 2004) and were extended by both authors for this study.

⁴Sizes were defined as greater than \$1 billion in average inflation-adjusted sales (large) (population is 19 firms), \$100 million to \$1 billion (medium) (population is 66 firms), and less than \$100 million (small) (population is 116 firms).

⁵Although we collapse these two types of entry modes in the core analysis to maximize the number of observations, we did investigate differences between the two entry modes. However, with only five companies entering through external means, it is difficult to say much statistically about this group. A qualitative review of the evidence suggests that there were not any systematic differences between the two groups.

⁶Three-digit optical patent classes are 356, 359, 372, and 385.

⁷Three-digit nonoptical telecom patent classes are 333, 340, 343, 367, 370, 375, 389, and 455.

⁸Because our constructs of interest are associated with relative emphasis in attention or orientation, these variables are entered as ratios. We did test all models with the numerators and denominators included separately. The results are robust to this alternative procedure, but the use of ratios is more consistent with our focus on emphasis.

⁹Every variable tested passes the Schoenfeld test of the Cox model’s inherent proportionality assumption. Thus, for any two observations with different levels of each time-varying covariate, the ratio of the two hazard functions is not dependent on time.

¹⁰All tests of economic significance are based on the full model (Model 6). In each case, the effect of the variable is evaluated at the median value and compared with the effect of the variable one standard deviation above the median.

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